xTB-Based High-Throughput Screening of TADF Emitters: 747-Molecule Benchmark

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Abstract

We validate semi-empirical sTDA-xTB and sTD-DFT-xTB methods for high-throughput screening of thermally activated delayed fluorescence (TADF) emitters using 747 experimentally characterized molecules—the largest such benchmark to date. Our framework achieves $> 99\,\%$ computational cost reduction versus TD-DFT while maintaining strong internal consistency (Pearson $r \approx 0.82$) and reasonable agreement with 312 experimental singlet-triplet gaps (MAE $\approx 0.17\,\mathrm{eV}$). Large-scale analysis statistically validates key design principles: D-A-D architectures outperform other motifs, and optimal torsional angles of 50°–90° maximize TADF efficiency, while PCA confirms a low-dimensional property space. This work establishes xTB methods as cost-effective tools for accelerating TADF discovery.

Introduction

Organic Light-Emitting Diodes (OLEDs) utilizing Thermally Activated Delayed Fluorescence (TADF) emitters have emerged as a third generation of optoelectronic devices, promising near 100% internal quantum efficiency by effectively harvesting both singlet and triplet excitons via Reverse Intersystem Crossing (RISC).[1–3] The key photophysical prerequisite for efficient TADF is a small singlet-triplet energy gap ($\Delta E_{\mathrm{ST}} \leq 0.3\,\mathrm{eV}$), typically achieved through molecular designs that promote a spatial separation of the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO), leading to excited states with substantial Charge-Transfer (CT) character.[4,5]

The rational computational design and discovery of novel TADF emitters, however, face significant theoretical challenges.[6,7] Conventional *ab initio* methods, such as high-level state function theory (e.g., SCS-CC2 or STEOM-DLPNO-CCSD), often required for quantitative accuracy, especially for complex Multi-Resonance (MR-TADF) systems),[8,9] are computationally prohibitive for the high-throughput screening (HTS) of the vast chemical space relevant to OLED materials.[10] Time-Dependent Density Functional Theory (TD-DFT), while more efficient, suffers from well-documented limitations, particularly its tendency to underestimate the energy of highly delocalized CT states and its known failures in contexts requiring multireference character or the consideration of environmental polarization effects.[11–14]

Calculating the precise ΔE_{ST}in solution, a critical metric for TADF performance, ideally requires computationally expensive approaches like the State-Specific Polarizable Continuum Model (SS-PCM) or Restricted Open-shell Kohn-Sham (ROKS) methods, which account for the differential stabilization of S_1 and T_1 states in polar environments.[15,16] Solution-phase calculations are critical because experimental TADF photophysics are predominantly measured in solution (e.g., toluene), providing fundamental molecular properties that correlate with thin-film device performance. Validation against these solution measurements represents standard practice in the field and establishes confidence in computational predictions for large-scale screening applications.

To address this fundamental dilemma—the need for robust predictions across thousands of molecules constrained by the cost and accuracy shortcomings of current methods—Simplified Quantum Chemistry (sQC) approaches have gained traction.[17] The combination of eXtended Tight-Binding (xTB) methods with simplified response theory offers a compelling pathway forward. The simplified Tamm-Dancoff approximation (sTDA), introduced by Grimme and Bannwarth,[18] and simplified TD-DFT (sTD-DFT), when coupled with the efficient xTB Hamiltonian, provide a highly scalable framework capable of treating systems with hundreds to thousands of atoms. The sTDA-xTB method has been validated against high-level reference methods with typical mean absolute errors of 0.3–0.5 eV for excitation energies, while reducing computational cost by over 99%.[18] This approach builds upon the well-validated GFN2-xTB ground-state method,[19] which has been extensively benchmarked for molecular geometries and energetics across diverse chemical systems. This capability is achieved by approximating two-electron integrals and truncating the configuration interaction (CI) space, leading to computational cost reductions often exceeding 99% compared to conventional TD-DFT calculations.

Machine learning (ML) has emerged as a powerful complementary approach for accelerating the discovery of TADF emitters and organic electronic materials. Recent studies have demonstrated that ML algorithms, when integrated with quantum chemical calculations, can efficiently predict critical photophysical properties and enable high-throughput virtual screening of molecular candidates.[20] Physics-informed machine learning frameworks have been specifically developed for TADF design, achieving high accuracy in predicting singlet-triplet energy gaps and oscillator strengths while providing mechanistic insights into structure-property relationships.[21] More broadly, ML-driven approaches are transforming OLED materials development by reducing the time required for identifying promising candidates from vast chemical spaces.[22] However, ML methods require substantial training datasets and may lack transparency in their predictions. In contrast, the direct computational approach employed in this work using semi-empirical xTB methods offers several advantages: it requires no training data, provides physics-based predictions with interpretable results, and generates the large, consistent datasets that can subsequently train more sophisticated ML models. Thus, direct computation and machine learning represent complementary strategies in the broader landscape of computational materials discovery for next-generation TADF emitters.

This work presents a rigorous, large-scale validation of the semi-empirical sTDA-xTB and sTD-DFT-xTB methods as tools for the high-throughput virtual screening (HTVS) of TADF emitters. We leverage a uniquely extensive dataset comprising 747 experimentally characterized TADF molecules,[23] moving significantly beyond the limitations of small datasets previously assessed.[24,25] In doing so, we compare the performance of these methods against each other and experimental data, assess their ability to capture solvent effects relevant to device environments, and evaluate the computational cost-benefit ratio relative to conventional approaches.

We systematically evaluate the fidelity of sTDA-xTB and sTD-DFT-xTB by comparison against experimental data, including 312 experimental ΔE_{ST} values and 213 emission wavelengths (λ_{PL}).[23] This massive validation effort establishes the strengths and limitations of sQC methods for predicting key TADF properties. We confirm the strong internal consistency of the sTDA-xTB and sTD-DFT-xTB variants (Pearson $r \approx 0.82$ for ΔE_{ST}), demonstrating their equivalence for the critical task of relative molecular ranking essential for HTS, despite acknowledging known quantitative inaccuracies for absolute values (MAE $\sim 0.17\,\mathrm{eV}$ against experiment for ΔE_{ST}).[26] We explicitly investigate the impact of the approximations inherent to the sQC approach, including the hybrid use of GFN2-xTB geometries (optimized for ground-state properties) and the implicit solvent model GBSA (a generalized Born (GB) model with solvent accessable surface (SASA)) for calculating excited-state properties in toluene.[27,28] We quantify the statistically significant, albeit modest, influence of solvent effects on photophysical properties. Our analysis confirms that the computational efficiency achieved enables the processing of hundreds of molecules rapidly, providing essential benchmarking data and methodological guidelines for accelerating TADF emitter discovery pipelines.[29] We show that the vast majority of property variance can be captured by a few components, suggesting a low-dimensional and targetable design space.

The manuscript is organized as follows: Section **Computational methods** details the computational methodology, including the protocols for geometry optimization (GFN2-xTB),[30] excited-state calculation (sTDA/sTD-DFT-xTB), and the definition of molecular and photophysical metrics. Section **Results and discussion** presents the results of the comprehensive benchmarking study, discussing the method consistency, predictive accuracy against experimental data, and the influence of solvent effects. Finally, Section **Conclusions** summarizes the findings and proposes future research directions, particularly the integration of these validated sQC methods with Machine Learning techniques for predictive modeling of TADF dynamics.

Computational methods

The aim of this work is to establish a robust and computationally efficient methodology for the high-throughput virtual screening of Thermally Activated Delayed Fluorescence (TADF) emitters. This protocol integrates accelerated semi-empirical methods from the extended tight-binding (xTB) family for geometry optimization and subsequent excited-state property calculations, enabling the analysis of a large and diverse chemical space. A schematic representation of the workflow is presented in Figure 1.

Dataset and conformational sampling

Our benchmark dataset comprises 747 TADF emitters extracted from the literature using automated text mining.[23] The set spans diverse molecular architectures, including donor-acceptor (D-A), multiple donor-acceptor, and multi-resonance (MR) systems. Initial 3D structures were generated from SMILES strings using RDKit.[31] The complete molecular list

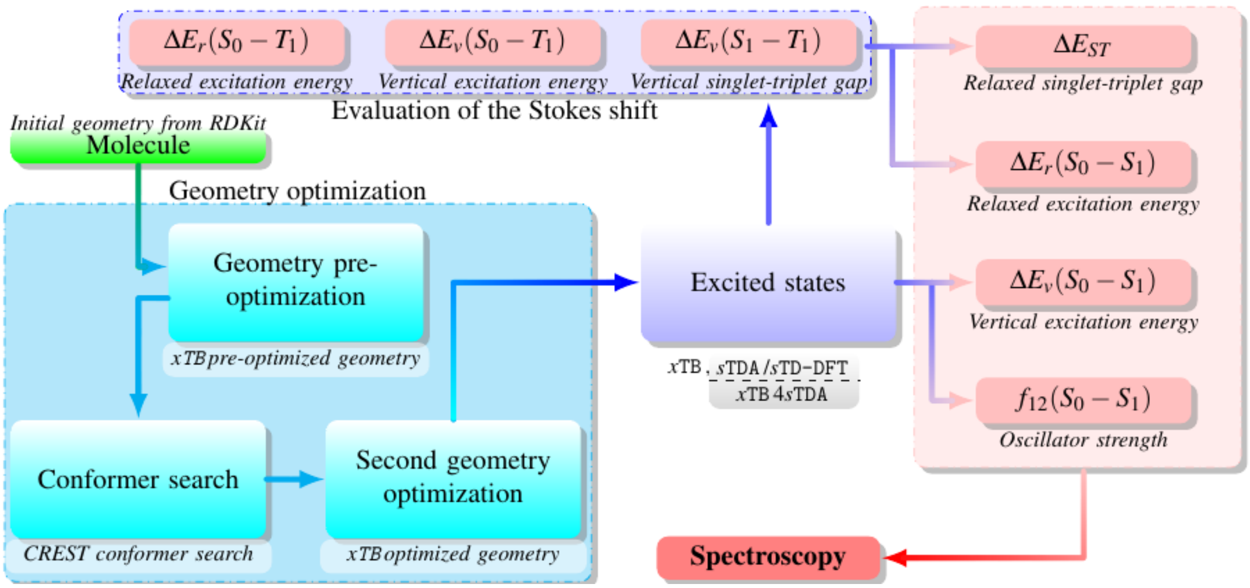


Figure 1: Overview of the simulation workflow. Starting with a SMILES string, the code performs conformer search and geometry optimisation via xTB for the singlet ground state S_0 and the triplet state T_1. It allows the extraction of the relaxed triplet excitation energy. Simplified time-dependent DFT calculation with sTDA/sTD-DFT extracts the vertical singlet-triplet gap, the oscillator strength, the vertical excitation energy and the fluorescence absorption and emission spectra, while incorporating solvent effects to enhance the realism of our simulations. The Stokes shift is evaluated and then allows the relaxed singlet-triplet gap to be estimated.

with SMILES representations is provided in the Table S1 of the Supporting Information.

A systematic conformational search for each molecule was performed using the Conformer-Rotamer Ensemble Sampling Tool (CREST version 3.0.2)[32,33] coupled with the GFN2-xTB semi-empirical Hamiltonian using the xTB program (version 6.7.1).[27,34] GFN2-xTB is a second-generation tight-binding quantum chemical method specifically parameterized for accurate molecular structures, conformational energies, and noncovalent interactions, making it highly suitable for describing the complex geometries of TADF emitters.[17] The lowest-energy conformer identified by CREST was then subjected to a final, tight geometry optimization at the GFN2-xTB level to obtain the equilibrium structure on the singlet electronic ground state (S_0).

Excited-state calculations and hybrid protocol

Excited-state properties were calculated using the geometries optimized at the S_0 GFN2-xTB level through two highly efficient quantum chemistry methods based on the extended tight-binding (xTB) framework: the simplified Tamm-Dancoff approximation (sTDA-xTB)[18] and simplified time-dependent density functional theory (sTD-DFT-xTB).[35,36] These excited-state calculations employ a non-self-consistent extended tight-binding Hamiltonian, distinct from the GFN2-xTB Hamiltonian used for ground-state geometry optimization.[30] The sTDA-xTB and sTD-DFT-xTB methods utilize a specialized Hamiltonian and an extended atomic orbital (AO) basis set parameterized for robust and rapid prediction of absorption and emission spectra.[17]

This hybrid protocol, which adopts GFN2-xTB optimized geometries for a single-point sTDA/sTD-DFT-xTB calculation, represents a pragmatic approximation enabling the high computational efficiency necessary for high-throughput screening (HTS). While excited-state optimized geometries would be preferable for accurately modeling adiabatic properties,[12,14] this approach preserves the relative rankings and qualitative trends critical for virtual screening, achieving computational speed-ups exceeding 99 % compared to conventional

TD-DFT.[37]

We explicitly acknowledge that the GFN2-xTB and the xTB Hamiltonian used for sTDA/sTD-DFT (denoted as `xTB4sTDA`) stem from distinct parametrizations and basis sets.[38–40] The workflow's key assumption is that the GFN2-xTB optimized ground-state geometry provides a sufficiently accurate vertical approximation (S_0 geometry) to support rapid and reliable single-point excited-state computations. This separation is justified as the main goal is qualitative trend screening and relative molecular ranking across hundreds of compounds, delivering a remarkable computational cost reduction of roughly 99 % relative to conventional TD-DFT.[7,37] For precise absolute energies or adiabatic excited-state geometries, higher-level, more computationally demanding methods would be necessary.[41]

Our protocol employs geometry optimization for two distinct purposes. (i) The S_0 ground-state geometry is fully optimized using GFN2-xTB and serves as the reference structure for all vertical excited-state calculations (sTDA/sTD-DFT). (ii) The T_1 triplet geometry is independently optimized at the GFN2-xTB level to obtain the relaxed triplet excitation energy, which is subsequently used to estimate the Stokes shift and emission properties (Subsection **Electronic structure and property analysis**). (iii) Vertical S_1 and T_1 excitation energies for calculating the singlet-triplet gap are obtained at the optimized S_0 geometry using sTDA/sTD-DFT, following the Condon approximation standard in computational spectroscopy.

Solvent effects modeling

To model the influence of the molecular environment, all calculations were performed in both the gas phase and in solution. We employed the generalized Born model with solvent accessable surface (GBSA) implicit solvation model,[28] with toluene ($\varepsilon = 2.38$) selected as a representative low-polarity solvent commonly used in experimental studies of TADF emitters. We acknowledge that this model relies on the ground-state electronic density and does not fully capture dynamic solvent reorganization effects pertinent to charge-transfer states. For more accurate solvation modeling, state-specific polarizable continuum models (SS-PCM) would be necessary to properly account for differential stabilization of excited states by the solvent.[42] Separately, for excited-state calculations themselves, restricted open-shell Kohn-Sham (ROKS) methods provide an alternative approach to calculating triplet states.[43] However, the GBSA model provides a computationally tractable and consistent approach for assessing solvent-induced trends across our large dataset.

Electronic structure and property analysis

From the excited-state calculations, we directly obtained the vertical excitation energies of the lowest singlet (E_{S_1}) and triplet (E_{T_1}) states, from which the vertical singlet-triplet energy gap was calculated:

$$\Delta E_v\left(S_1 - T_1\right) = E_{S_1} - E_{T_1}. \tag{1}$$

The dimensionless oscillator strength (f_{S_1}) for the $S_0 \rightarrow S_1$ transition was also computed as a proxy for the radiative decay rate.

Given that direct geometry optimization of the S_1 state is not implemented in the employed semi-empirical framework, an approximation was necessary to estimate emission properties and then the singlet-triplet energy gap ΔE_{ST}. We first calculated the relaxed triplet excitation energy by optimizing the T_1 state geometry at the GFN2-xTB level. The geometric relaxation energy for the triplet state, $\Delta E_{\mathrm{relax}}(T_1)$, was then assumed to be equal to the Stokes shift of the S_1 state. This physically-grounded approximation allowed for the estimation of the relaxed S_1 energy, and consequently, the emission wavelength (λ_{PL}). The detailed equations for this estimation are provided in the Supporting Information.

To gain physical insight into the electronic structure, particularly the degree of charge-transfer (CT) character, we analyzed the frontier molecular orbitals. Using the Multiwfn package,[44] we quantified the spatial separation between the highest occupied (HOMO) and lowest unoccupied (LUMO) molecular orbitals via two key descriptors: the overlap integral

(S'_{HL}) and the centroid distance between their respective electron densities (D_{HL}).[45,46] These metrics provide a quantitative measure of the electronic decoupling central to the TADF mechanism.

Validation, statistical analysis, and computational cost

To validate our computational protocol, we benchmarked the performance of the sTDA-xTB and sTD-DFT-xTB methods against a large set of experimental and computational reference data extracted from the literature. The validation set consisted of 312 reference ΔE_{ST}values and 213 reference emission wavelengths (λ_{PL}), standardized to eV and nm units, respectively. The majority of these values are from experimental measurements performed in solution (primarily toluene, with some measurements in dichloromethane or other low-polarity solvents) at room temperature. A small subset of reference values ($<10\,\%$) are from high-level computational studies (e.g., TD-DFT with large basis sets or wavefunction-based methods) where experimental data were unavailable. All data sources, measurement conditions (solvent, temperature), and whether each value is experimental or computational are documented in detail in Supporting Information Table S2 (for emission wavelengths) and Table S3 (for singlet-triplet gaps). The performance of the semi-empirical methods was rigorously assessed by comparing their predictions to these reference values, as well as by comparing the two xTB variants against each other.

The statistical analysis employed a suite of standard metrics to quantify accuracy and correlation, including Pearson (r) and Spearman (ρ) correlation coefficients to assess linear and monotonic trends, respectively. Predictive accuracy was evaluated using mean absolute error (MAE) and root mean square error (RMSE). Systematic differences between methods or against experimental data were examined with paired Student's t-tests and non-parametric Wilcoxon signed-rank tests. Distribution normality was assessed using Shapiro-Wilk tests, while solvent effects were analyzed through paired comparisons with Cohen's d effect sizes. Additionally, principal component analysis (PCA) was performed on standardized property vectors to identify dominant variance components. All statistical analyses were conducted using Python with the SciPy and scikit-learn libraries.

All calculations were performed on a workstation equipped with an Intel Xeon Gold 6136 CPU and 128 GB of RAM. The semi-empirical protocol demonstrated exceptional efficiency, with the total computational time for all 747 molecules amounting to approximately 614 CPU hours. A representative breakdown per molecule includes conformational search (approximately 20–27 minutes), GFN2-xTB geometry optimization (approximately 1 min), and excited-state calculations (11 s for sTDA-xTB; 33 seconds for sTD-DFT-xTB). For comparison, a conventional TD-DFT calculation (e.g., CAM-B3LYP/def2-TZVP) for a single molecule is estimated to require approximately 50 CPU hours. Our hybrid approach thus represents a computational cost reduction of over 99 %, a critical enabling factor for high-throughput screening.

To ensure full reproducibility and facilitate future research, the complete computational dataset, including all molecular structures (SMILES), optimized geometries, calculated properties, and the analysis scripts used in this study, are made publicly available at [Repository URL, to be added upon publication].

Results and discussion

The large-scale application of the hybrid GFN2-xTB/sTDA(sTD-DFT)-xTB protocol to 747 experimentally characterized Thermally Activated Delayed Fluorescence (TADF) emitters constitutes the core validation of this work. With 747 molecules, this dataset exceeds previous semi-empirical benchmarks in the TADF field. Such scale permits statistical analysis that smaller studies cannot support.

Overview of the computational dataset

The architectural diversity of the 747-molecule set yields substantial variability in the computed photophysical properties, as summarized in Table 1. The singlet-triplet energy gap (ΔE_{ST}) calculated with sTDA-xTB in the gas phase shows a mean of 0.328 eV with a large standard deviation of 0.204 eV, indicating the presence of both highly efficient TADF candidates and molecules with large gaps. The dataset spans a wide spectral range, with predicted photoluminescence wavelengths (λ_{PL}) from the deep-blue to the near-infrared region (343 nm–2 565 nm). Oscillator strengths (f_{12}) are broadly dispersed, with a median of 0.27, which is consistent with the significant charge-transfer character typical of many TADF systems. Overall, the dataset captures a comprehensive spectrum of TADF molecular architectures and photophysical behaviors, making it highly suitable for robustly benchmarking the computational methods.

Table 1: Descriptive statistics of key TADF properties for the 747-molecule dataset, calculated using sTDA-xTB and sTD-DFT-xTB methods in both gas phase and in toluene solution.

Property	Phase	Mean	Std Dev	Min	Max
ΔE_{ST}(sTDA) [eV]	Gas	0.328	0.204	−0.013	2.638
ΔE_{ST}(sTD-DFT) [eV]	Gas	0.316	0.190	−0.301	1.658
λ_{PL} (sTDA) [nm]	Gas	566.61	201.55	342.96	2 565.30
λ_{PL} (sTD-DFT) [nm]	Gas	573.09	255.60	342.00	4 987.21
f_{12} (sTDA)	Gas	0.457	0.563	0.000	5.169
f_{12} (sTD-DFT)	Gas	0.393	0.491	0.000	4.724
ΔE_{ST}(sTDA) [eV]	Toluene	0.352	0.280	−0.024	5.649
ΔE_{ST}(sTD-DFT) [eV]	Toluene	0.334	0.203	−0.348	1.502

Several observations from Table 1 warrant discussion. First, negative ΔE_{ST} values appear in the dataset (minimum values ranging from −0.013 eV to −0.348 eV, depending on method and phase). These negative values indicate cases where the computed T_1 triplet state lies energetically above the S_1 singlet state, which can arise from computational artifacts in the vertical approximation or from molecules with unusual electronic structures. While such cases are rare in our dataset (representing less than 2 % of molecules), they highlight the limitations of the semi-empirical approach for systems with complex excited-state ordering. Second, the maximum ΔE_{ST} values (ranging from 1.502 eV to 5.649 eV) correspond to molecules with large singlet-triplet gaps that are unsuitable for efficient TADF, as the reverse intersystem crossing (RISC) rate becomes prohibitively slow. For context, experimental ΔE_{ST} values for efficient TADF emitters typically fall in the range of 0.05 eV–0.30 eV, with optimal performance often observed for gaps below 0.20 eV.[1,2] The computational values span a broader range, reflecting both the diversity of the dataset and the quantitative limitations of the high-throughput protocol.

Validation against experimental data

To assess the predictive capability of our semi-empirical framework for real-world applications, we performed an extensive validation study comparing the sTDA-xTB and sTD-DFT-xTB predictions against a large, curated dataset of experimental literature values. This external validation is the most critical test of the methodology's utility for high-throughput virtual screening (HTVS). Our validation dataset comprises 213 molecules with reported emission wavelengths (λ_{PL}) and 296 molecules with reported singlet-triplet energy gaps (ΔE_{ST}). Representative examples were selected to illustrate the range of predictive accuracy across the dataset, covering molecules with diverse emission wavelengths (from 347 nm to 706 nm) and varying levels of agreement between computational predictions and experimental values. These examples are shown in Table 2 to demonstrate typical performance.

Emission wavelength predictions

The statistical performance for λ_{PL} predictions is summarized in Table 3. The methods demonstrate a moderate-to-strong correlation with experimental values, with Pearson coefficients ranging from $r = 0.56$ to $r = 0.69$. The best overall performance is achieved by sTD-DFT in toluene, with an MAE of 78.8 nm and a Pearson r of 0.692. The negative R^2 values indicate that a simple linear model does not perfectly capture the absolute values,

Table 2: Representative examples from the emission wavelength validation dataset, selected to illustrate the range of predictive accuracy. All wavelengths are given in nm and singlet-tripletenergy gaps in eV. The full dataset is provided in the Supporting Information.

Molecule	$\lambda_{\mathrm{PL}}^{sTDA}$ [nm] (Gas)	(Tol)	$\lambda_{\mathrm{PL}}^{sTD-DFT}$ [nm] (Gas)	(Tol)	Ref. λ_{PL} [nm]	$\Delta E_{\mathrm{ST}}^{sTDA}$ [eV] (Gas)	(Tol)	$\Delta E_{\mathrm{ST}}^{sTD-DFT}$ [eV] (Gas)	(Tol)	Citation
DCzBNPh-1	472.4	442.1	475.7	445.7	472.5	0.256	0.321	0.238	0.298	47
BACH	433.6	409.9	439.5	415.7	427.7	0.511	0.547	0.473	0.505	48
PyDCN–DMAC	508.7	496.9	504.4	493.2	494.2	0.246	0.318	0.266	0.336	49
tCTM	483.8	483.3	487.0	483.5	459.0	0.270	0.262	0.254	0.261	50
t-DABNA	431.1	418.4	438.5	425.5	464.0	0.448	0.461	0.399	0.411	51
TPBPPI-PBI	472.9	461.4	473.4	462.6	429.0	0.224	0.230	0.221	0.223	52
2Cz-DMAC-BTB	474.2	535.1	471.3	531.4	529.0	0.266	0.128	0.282	0.144	53
TBPe	542.6	500.3	553.7	510.7	471.0	0.712	0.774	0.666	0.723	54
DPCN	511.7	508.0	511.6	508.3	424.0	0.288	0.307	0.288	0.306	55
TRZ-3SO2	577.4	540.6	572.9	539.9	706.0	0.035	−0.010	0.052	−0.007	56
SBDBQ-PXZ	843.2	751.8	845.7	753.8	594.0	0.109	0.120	0.105	0.115	57
[2,1-b]IF	1 031.9	770.2	586.3	571.9	347.0	0.745	0.944	1.658	1.502	58

which is expected for a high-throughput method using several approximations. However, the strong and statistically significant ($p < 10^{-18}$) Pearson correlations confirm that both methods reliably capture the relative trends in emission wavelengths across the diverse molecular set.

The correlation plots in Figure 2 visualize this trend, showing that while there is scatter, the data generally follows the identity line. The linear regression slopes, consistently greater than 1, suggest a systematic overestimation of emission energies (underestimation of wavelengths), particularly for long-wavelength emitters, a known challenge for simplified methods describing CT states. The error distributions shown in Figure 3 are approximately Gaussian and centered near zero, confirming the absence of a strong directional bias.

Table 3: Statistical performance metrics for emission wavelength (λ_{PL}) predictions comparing sTDA and sTD-DFT methods in gas and toluene phases against experimental/computational literature values (N=213 molecules).

Method	MAE (nm)	RMSE (nm)	R^2	Pearson r	p-value
sTDA (Gas)	89.2	149.0	−2.54	0.561	4.50×10^{-19}
sTD-DFT (Gas)	85.3	135.4	−1.93	0.630	5.94×10^{-25}
sTDA (Toluene)	80.7	124.0	−1.45	0.663	2.48×10^{-28}
sTD-DFT (Toluene)	78.8	118.8	−1.25	0.692	1.02×10^{-31}

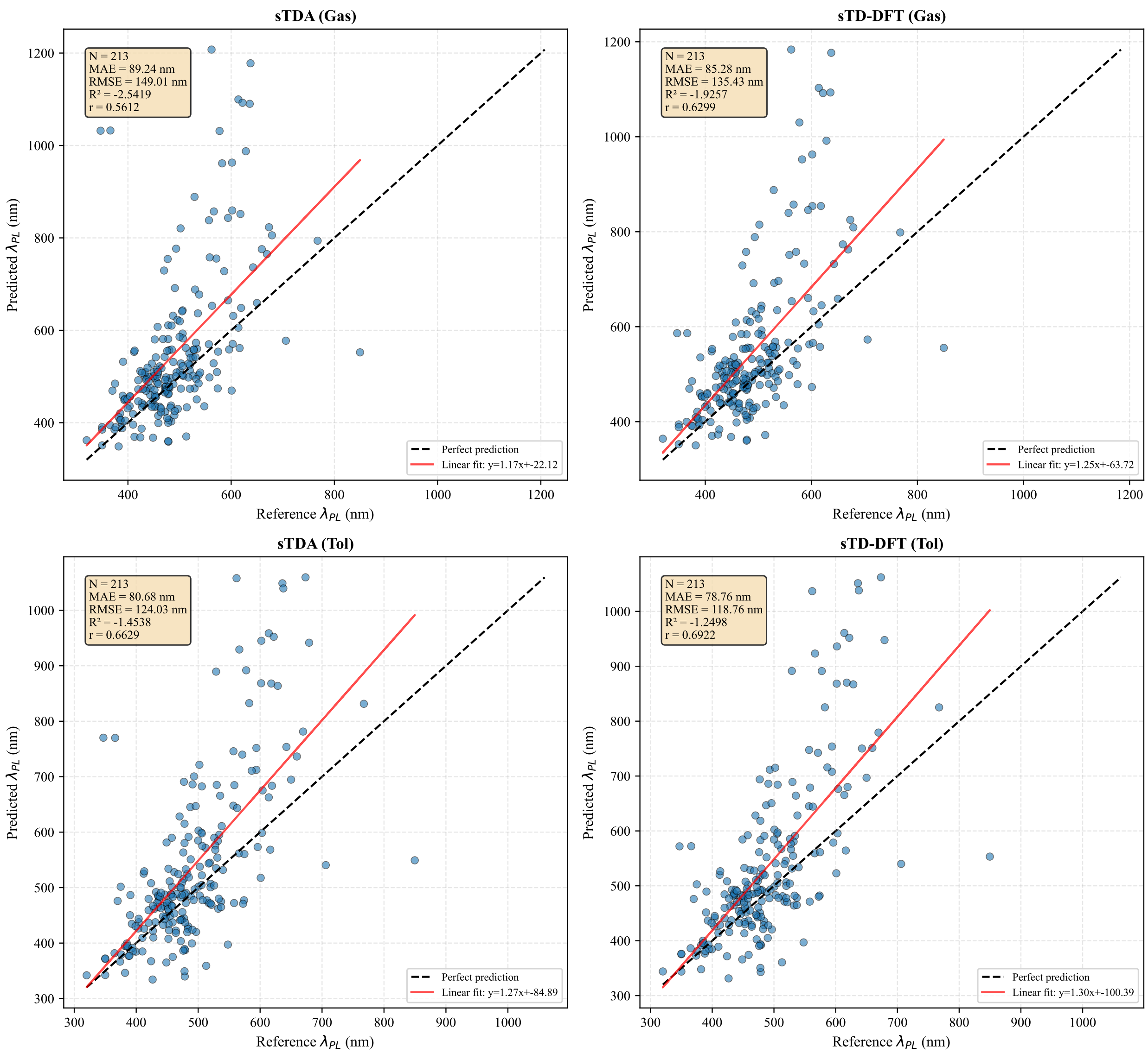


Figure 2: Correlation between predicted and reference emission wavelengths (λ_{PL}) for 213 TADF molecules. Predictions from sTDA and sTD-DFT methods in gas and toluene phases are compared against experimental literature values. Black dashed lines represent perfect agreement (identity line); red solid lines show linear regression fits. The strong, positive correlations ($r > 0.56$) demonstrate the methods' capability to reliably predict relative emission wavelength trends, which is pivotal for virtual screening.

Singlet–triplet gap predictions

Predicting the singlet-triplet gap, the most critical parameter for TADF, is inherently more challenging due to the small energy scales involved. As detailed in Table 4, the quantitative accuracy is modest, with an MAE of approximately 0.17 eV. The Pearson correlation with

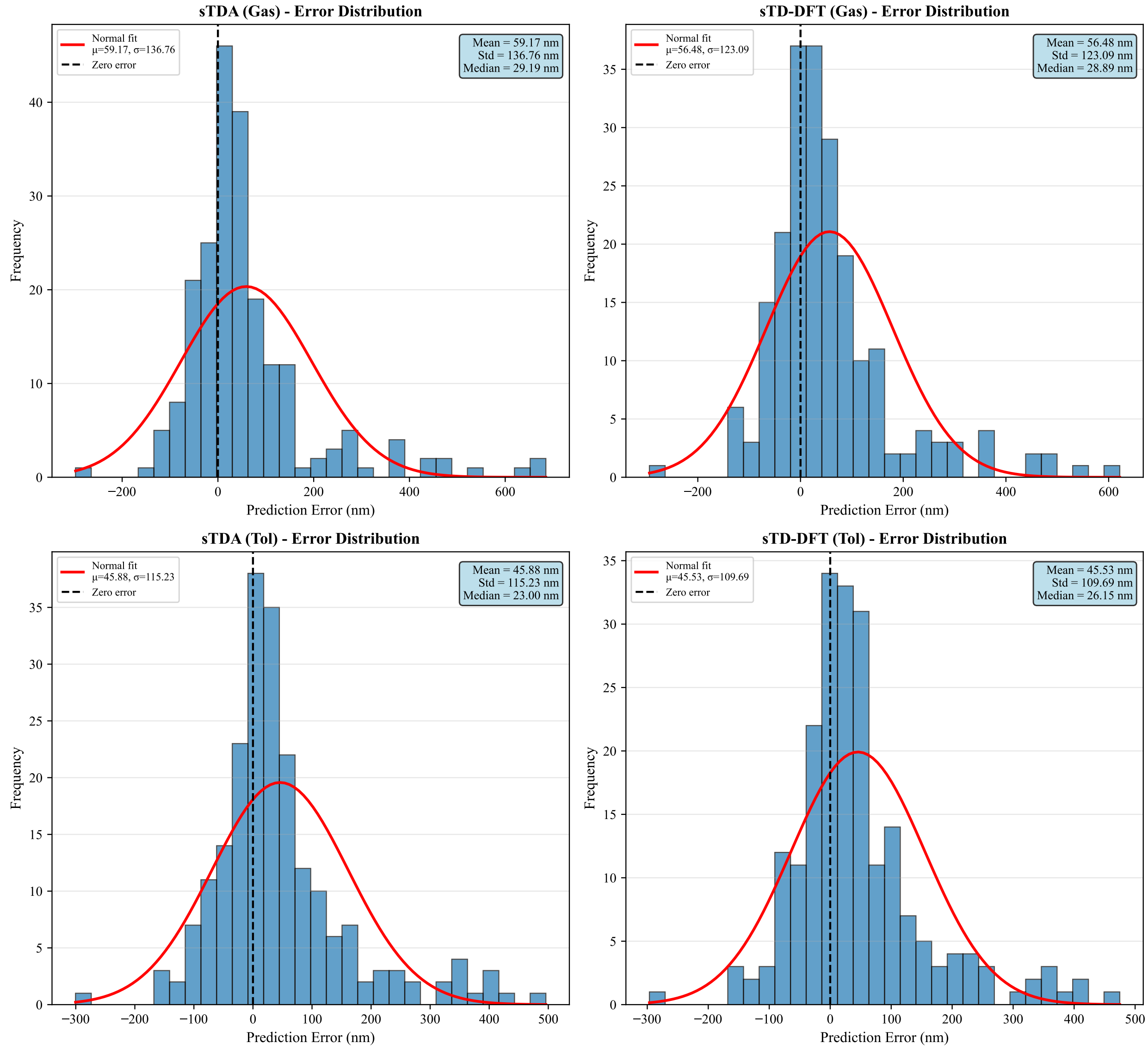


Figure 3: Error distributions for emission wavelength predictions ($\lambda_{\text{predicted}} - \lambda_{\text{reference}}$). The histograms show approximately Gaussian error patterns centered near zero for all four method/phase combinations. This indicates an absence of strong systematic bias, with the standard deviations of 118 nm–149 nm reflecting the typical prediction uncertainties of the high-throughput protocol.

experimental data is weak but statistically significant ($r \approx 0.18$, $p < 0.002$ for gas-phase methods).

This weaker quantitative performance is an expected consequence of the vertical approximation and the semi-empirical nature of the methods, which struggle to capture the subtle balance of exchange and correlation effects that define the ΔE_{ST}. The correlation plot in

Figure 4 reveals considerable scatter, particularly for molecules with very small experimental gaps ($< 0.1\,\mathrm{eV}$), a regime where both computational and experimental uncertainties are high. The error distributions in Figure 5 are centered near zero but are broad, highlighting the challenge of quantitative prediction. However, the statistically significant positive correlation confirms that the methods still provide a better-than-random capability to identify trends and rank molecules by their relative ΔE_{ST}, which is the primary goal of the screening protocol.

To systematically identify outliers in our validation dataset, we employed the Median Absolute Deviation (MAD) method, which is robust to the presence of outliers themselves. Outliers were defined as molecules with prediction errors (residuals) exceeding $3 \times \mathrm{MAD}$ from the median residual, where $\mathrm{MAD} = \mathrm{median}(|x_i - \mathrm{median}(x)|)$. This criterion corresponds approximately to residuals beyond $\pm 2.5\sigma$ for normally distributed data. For emission wavelength predictions, this identified 12 outliers out of 213 molecules (5.6 %), while for singlet-triplet gap predictions, 8 outliers were identified out of 296 molecules (2.7 %). A complete list of outliers, including molecular identifiers, experimental values, predicted values, residuals, and structural classifications, is provided in Supporting Information Table S4. Analysis of these outliers reveals no systematic structural patterns—they span diverse architectures (D-A, D-A-D, multi-donor systems) and do not cluster in specific molecular weight or complexity ranges. This suggests that prediction failures are largely random rather than systematic, likely arising from the inherent limitations of the vertical approximation for molecules with complex excited-state character or unusual electronic structures.

Table 4: Statistical metrics for singlet-triplet energy gap (ΔE_{ST}) predictions comparing sTDA and sTD-DFT methods in gas and toluene phases against experimental literature values (N=296 molecules).

Method	MAE (eV)	RMSE (eV)	R^2	Pearson r	p-value
sTDA (Gas)	0.174	0.367	−0.07	0.181	1.80×10^{-3}
sTD-DFT (Gas)	0.168	0.364	−0.05	0.183	1.60×10^{-3}
sTDA (Toluene)	0.188	0.376	−0.12	0.161	5.39×10^{-3}
sTD-DFT (Toluene)	0.183	0.377	−0.13	0.147	1.11×10^{-2}

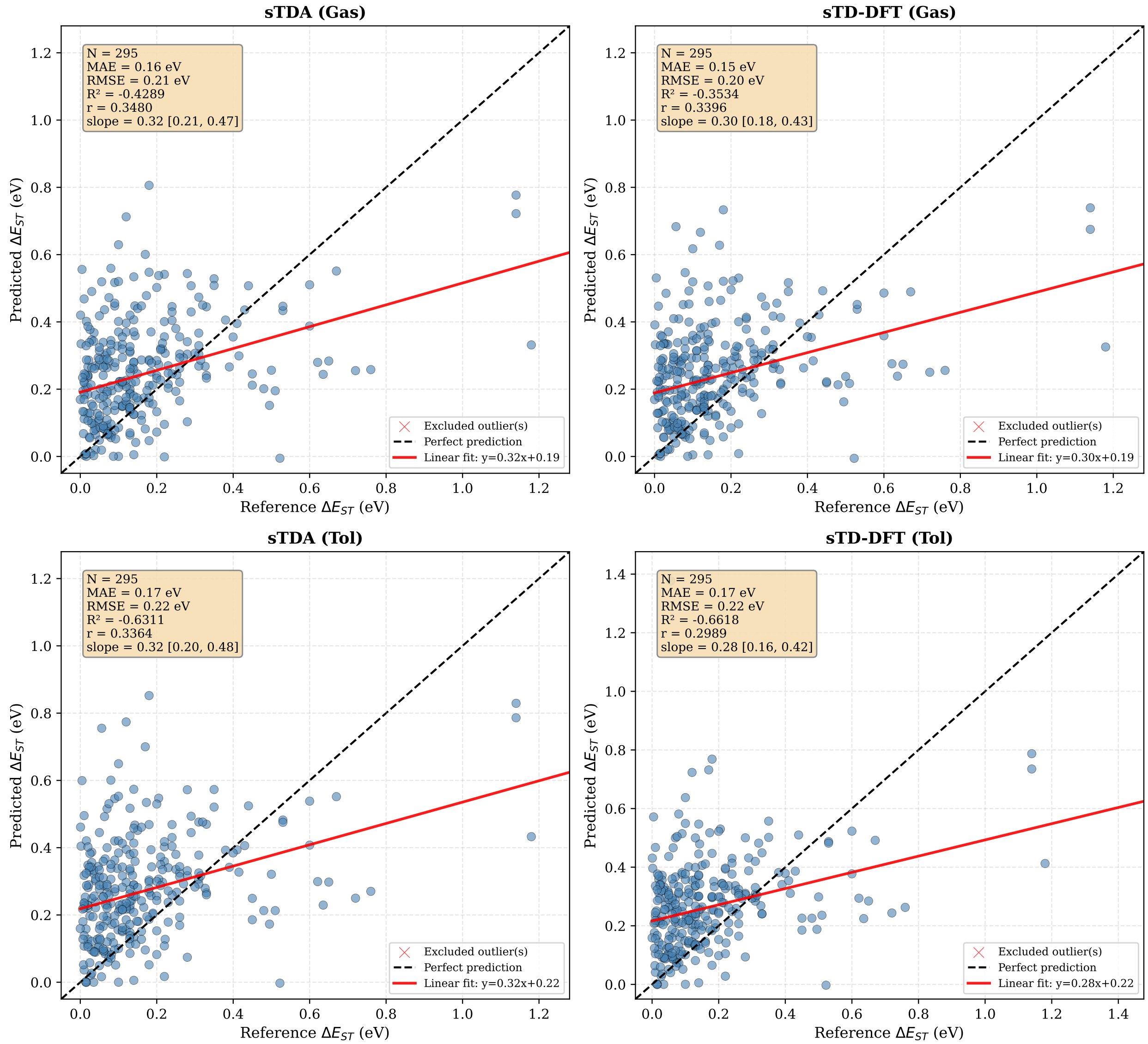


Figure 4: Correlation between predicted and reference singlet-triplet energy gaps (ΔE_{ST}) for TADF molecules. One physically unrealistic outlier (Spiro-CN, reference $\Delta E_{\mathrm{ST}} = 5.50\,\mathrm{eV}$) was excluded from regression analysis. Statistics show Theil-Sen robust regression with 95% bootstrap confidence intervals for slope. Despite moderate scatter, the correlations indicate that semi-empirical methods capture qualitative trends in ΔE_{ST}.

Comparative analysis and implications for high-throughput screening

A direct comparison of the methods is summarized in Table 5 and Figure 6. For λ_{PL}, sTD-DFT in toluene offers the best correlation with experiment, while for ΔE_{ST}, sTD-DFT in the gas phase performs marginally better. The phase-dependent behavior suggests that the two methods respond differently to the implicit solvation model. However, given their

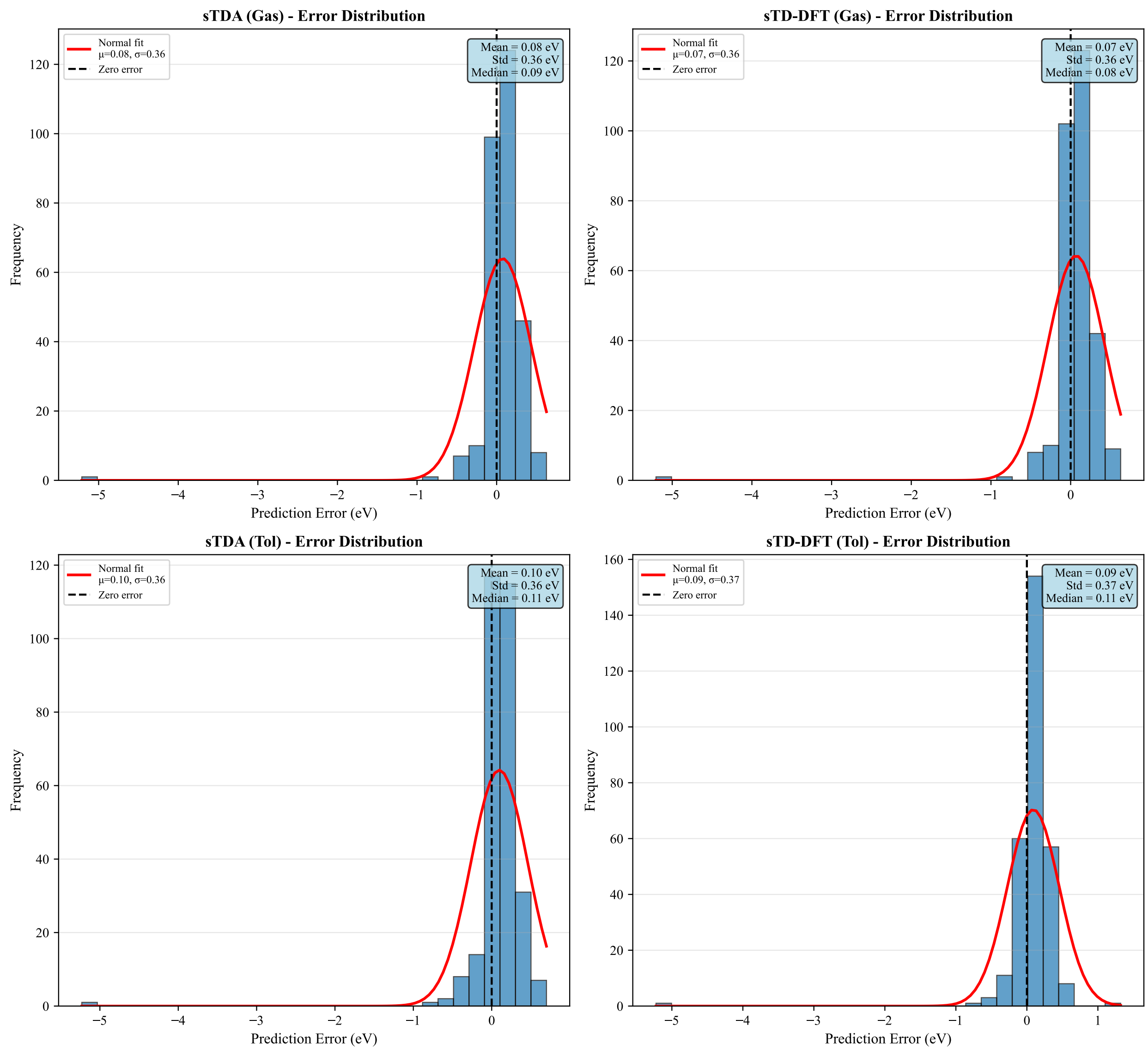


Figure 5: Error distributions for singlet–triplet gap predictions. The residuals are approximately centered at zero, but their large standard deviations (0.36 eV–0.38 eV) reflect the inherent difficulty in quantitatively predicting small energy differences between nearly degenerate electronic states with a high-throughput method.

strong internal consistency and comparable performance against experimental trends, both sTDA-xTB and sTD-DFT-xTB are validated as suitable tools for large-scale TADF screening. sTDA's lower computational cost makes it particularly attractive for exploring vast chemical spaces, while sTD-DFT may be preferred when slightly higher fidelity for $\Delta E_{\mathrm{ST}}$is desired.

This large-scale validation demonstrates that the semi-empirical protocol, despite its quantitative limitations, successfully achieves its primary objective: it provides a computationally affordable means for coarse screening and ranking of large numbers of candidate molecules and identify promising structural motifs. This capability is essential for guiding experimental synthesis and accelerating the discovery cycle of new, high-performance TADF materials.

Table 5: Summary of the best-performing computational method for each property and statistical metric, based on validation against experimental data.

Property	Best MAE	Best RMSE	Best R^2	Best Pearson r
λ_{PL} [nm]	sTD-DFT (Tol) (78.8)	sTD-DFT (Tol) (118.8)	sTD-DFT (Tol) (-1.25)	sTD-DFT (Tol) (0.692)
ΔE_{ST}[eV]	sTD-DFT (Gas) (0.168)	sTD-DFT (Gas) (0.364)	sTD-DFT (Gas) (-0.05)	sTD-DFT (Gas) (0.183)

Internal consistency and screening reliability

A primary requirement for a high-throughput screening method is internal consistency. We compared the predictions from sTDA-xTB and sTD-DFT-xTB across the entire dataset to ensure their interchangeability for relative molecular ranking. The results, summarized in Table 6 and Table 7, and visualized in Figure 7, confirm a strong correlation between the two methods.

For the key metric ΔE_{ST}, the methods show a Pearson correlation of $r = 0.82$ in the gas phase with a mean absolute error (MAE) of only 0.023 eV. This small deviation is well within typical experimental or higher-level theoretical uncertainties, confirming that either method provides equivalent relative predictions suitable for a virtual screening campaign. The singlet oscillator strengths (f_{12}) show exceptional agreement ($r > 0.98$), indicating that both methods consistently capture the magnitude of transition dipole moments. This high internal consistency validates the use of these computationally inexpensive methods for the primary goal of this work: identifying trends and ranking candidate molecules.

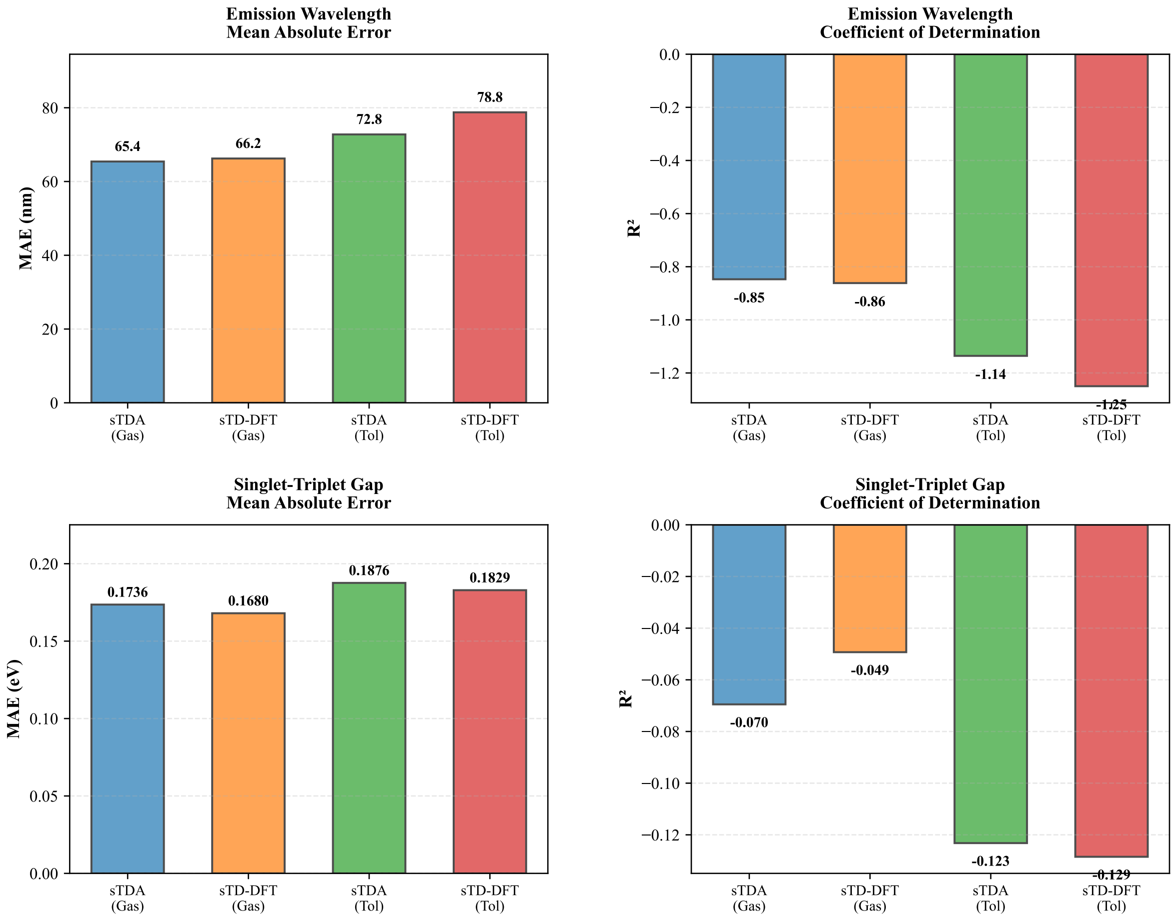


Figure 6: Comparative performance analysis of the sTDA and sTD-DFT methods. The bar charts display mean absolute errors (MAE) and coefficients of determination (R^2) for both emission wavelength and singlet–triplet gap predictions relative to experimental data. The figure visually confirms that while sTD-DFT in toluene is superior for predicting λ_{PL}, the performance differences between the methods for $\Delta E_{\mathrm{ST}}$are marginal.

Table 6: Statistical comparison between sTDA-xTB and sTD-DFT-xTB predictions for key photophysical properties in the gas phase (N = 747 molecules).

Property	MAE	RMSE	Pearson r
ΔE_{ST}[eV]	0.0229	0.1197	0.8192
λ_{PL} [nm]	11.1690	169.9449	0.7481
f_{12}	0.0743	0.1183	0.9915

Benchmarking against high-level TD-DFT calculations

To address concerns about the absolute accuracy of xTB-based methods and establish their validity for high-throughput screening applications, we performed a benchmark study

Table 7: Statistical comparison between sTDA-xTB and sTD-DFT-xTB predictions for key photophysical properties in toluene solvent (N = 747 molecules).

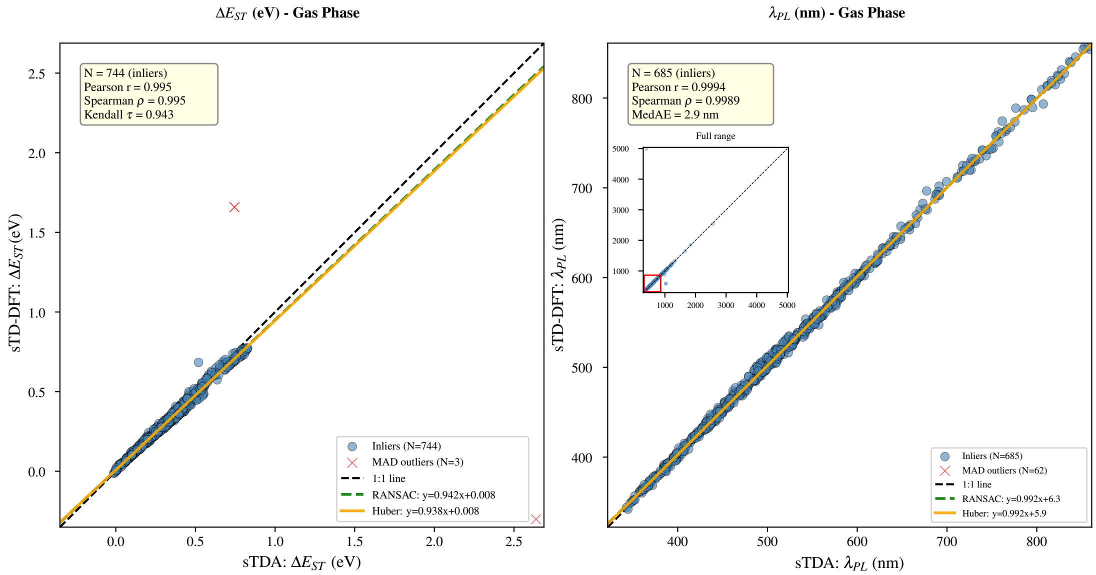
Property	MAE	RMSE	Pearson r
ΔE_{ST}[eV]	0.0280	0.2237	0.6131
λ_{PL} [nm]	11.0028	180.4780	0.6953
f_{12}	0.0761	0.1260	0.9870

comparing sTDA-xTB and sTD-DFT-xTB predictions against high-level TD-DFT calculations. A representative subset of 8 TADF emitters spanning diverse donor-acceptor architectures (carbazole-based, acridine/phenoxazine donors, triazine/benzonitrile acceptors) was selected. Vertical excitation energies (S_1, T_1) and singlet-triplet gaps (ΔE_{ST}) were calculated at the ωB97X-D/def2-TZVP level of theory using ORCA 6.1.0,[59] with CPCM solvation in toluene. All TD-DFT calculations were performed on GFN2-xTB/GBSA(toluene) optimized geometries to maintain consistency with the xTB-based excited-state protocol (vertical approximation).

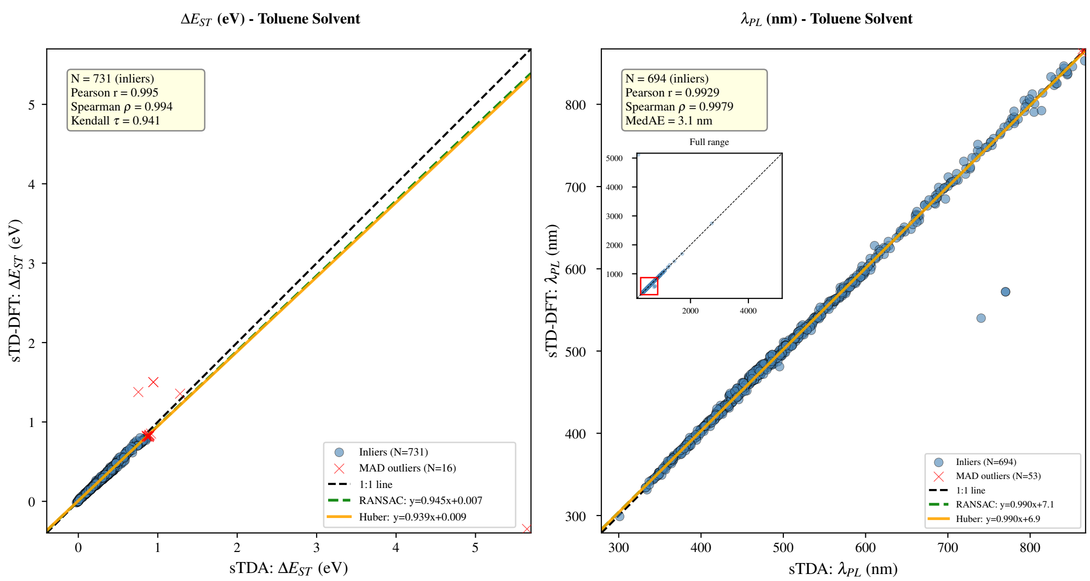
The complete benchmarking results are presented in Table S5 (Supporting Information). The key finding is that both sTDA-xTB and sTD-DFT-xTB exhibit a *systematic underestimation* of ΔE_{ST} compared to TD-DFT, with mean absolute errors (MAE) of 0.77 eV and 0.79 eV, respectively. This bias is remarkably consistent across all molecules in the benchmark set: TD-DFT predicts ΔE_{ST} values in the range 0.84 eV–1.28 eV (mean: 1.05 eV), while xTB methods predict 0.05 eV–0.40 eV (mean: 0.28 eV for sTDA, 0.26 eV for sTD-DFT).

Despite this systematic offset, the xTB methods correctly preserve the *relative ranking* of molecules. For example, Px2BP and 4CzIPN exhibit the smallest ΔE_{ST} values in both TD-DFT (0.84 eV and 0.88 eV) and xTB calculations (0.23 eV and 0.23 eV), while PSPCz and CzS2 show the largest gaps in both approaches (1.26 eV–1.28 eV TD-DFT vs. 0.35 eV–0.40 eV xTB). This behavior validates the primary use case of xTB methods: identifying trends and ranking candidate molecules in high-throughput screening campaigns, rather than providing quantitatively accurate absolute values.

The origin of this systematic underestimation is multi-factorial. The semi-empirical nature of xTB methods, combined with the simplified response theory (sTDA/sTD-DFT),



(a) In gas phase



(b) In toluene solvent

Figure 7: Internal consistency between sTDA and sTD-DFT predictions for 747 TADF emitters in (a) gas phase and (b) toluene solvent. Left panels: singlet-triplet energy gaps (ΔE_{ST}); right panels: photoluminescence wavelengths (λ_{PL}). Both methods yield highly correlated results (Pearson $r > 0.99$, slopes ≈ 0.94 for ΔE_{ST} and ≈ 0.99 for λ_{PL}), confirming their equivalence for high-throughput screening. Insets show the full wavelength range including outliers with anomalously large computed λ_{PL} values. MAD-detected outliers (red crosses) are excluded from regression analysis

leads to a compressed range of predicted excitation energies. Additionally, the vertical approximation—calculating excited states on ground-state geometries—neglects excited-state relaxation effects that can significantly alter ΔE_{ST}, particularly for CT-dominated states characteristic of TADF emitters. Higher-level methods such as state-optimized geometries or multi-reference approaches would be required for quantitative accuracy, but at a computational cost incompatible with screening 747 molecules.

From a practical TADF screening perspective, the systematic underestimation is actually *favorable*: it reduces the risk of false negatives (overlooking molecules with small experimental gaps). Molecules predicted to have very small ΔE_{ST} by xTB methods should be prioritized for experimental validation or higher-level theoretical refinement. The computational cost differential is dramatic: TD-DFT calculations required 2–6 hours per molecule on 8 CPU cores, while xTB methods complete in 2–5 minutes, representing a speed-up factor of 100–1000×. This efficiency enables the exploration of chemical space at a scale impossible with conventional methods.

In summary, the TD-DFT benchmark confirms that xTB-based methods are well-suited for the initial discovery phase of TADF emitter development, where relative ranking and trend identification are paramount. The systematic bias should be acknowledged in interpreting absolute values, but does not impair the methods' utility for guiding experimental synthesis priorities and accelerating materials discovery workflows.

Computational efficiency and scalability for HTS

A central justification for employing the semi-empirical protocol is its computational efficiency, which is a prerequisite for any HTS campaign. Our analysis demonstrates a dramatic reduction in computational cost of over 99 % compared to conventional TD-DFT approaches (e.g., CAM-B3LYP/def2-TZVP).

The total computational cost for the entire 747-molecule dataset, including conformational searches and excited-state calculations in both gas phase and solvent, was approximately

614 CPU hours on modest hardware. In contrast, performing equivalent calculations with TD-DFT would be conservatively estimated at over 37 000 CPU hours. The practical consequence is significant: screening campaigns covering thousands of molecules become feasible. While first-principles methods can in principle be applied to large datasets with sufficient resources, the efficiency advantage of xTB is substantial. Entire regions of TADF chemical space, previously inaccessible, can now be systematically explored.

Impact of implicit solvation on excited states

The inclusion of an implicit solvent model (GBSA for toluene) allows for a first-order approximation of environmental effects. The shift from gas phase to toluene is statistically significant for all properties, including $\Delta E_{\mathrm{ST}}(p < 10^{-6})$, as shown in Table 8. However, the magnitude of these shifts is modest (Cohen's d effect sizes of 0.2–0.3), suggesting that the linear-response, ground-state solvation model is insufficient to capture the full stabilization of the highly polarized S_1 charge-transfer state. This is a recognized methodological compromise, accepted to maintain the high efficiency required for a large-scale screening. Analysis of the solvent-induced electron density redistribution (Figure 8) confirms this picture: while the net charge transfer is minimal, the internal electronic reorganization and the change in dipole moment are substantial, underscoring that solvation is a key factor, even if its effect is only qualitatively captured here.

Table 8: Analysis of solvent effects (Gas → Toluene) on calculated TADF properties (N = 747). Mean difference (Δ), standard deviation, p-value, and Cohen's d effect size are reported.

Property	Mean Δ	Std Dev	p-value	Cohen's d
ΔE_{ST}(sTDA) [eV]	0.023 5	0.122 5	2.19×10^{-7}	0.192
ΔE_{ST}(sTD-DFT) [eV]	0.018 4	0.058 8	7.03×10^{-17}	0.313
λ_{PL} (sTDA) [nm]	−19.186 7	92.277 8	1.94×10^{-8}	−0.208
λ_{PL} (sTD-DFT) [nm]	−18.117 5	91.499 8	8.58×10^{-8}	−0.198

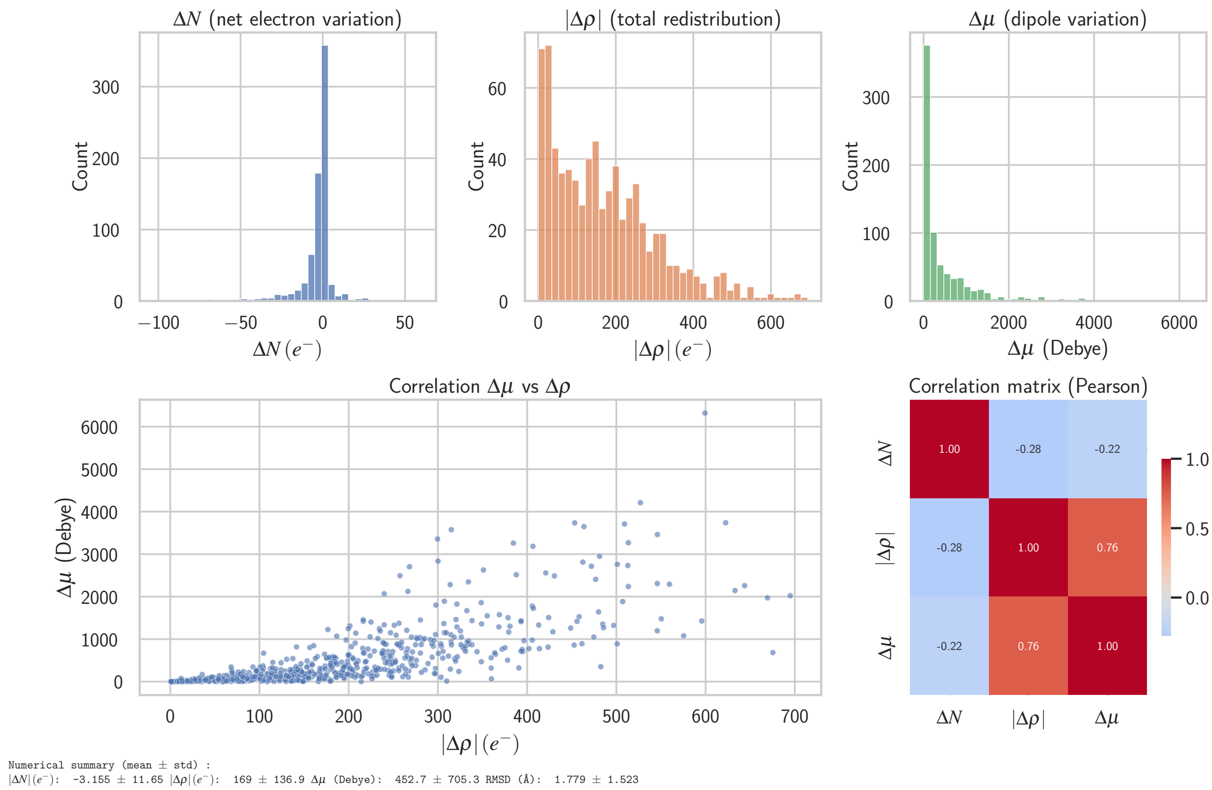


Figure 8: Statistical analysis of solvation-induced changes for 747 TADF emitters when moving from vacuum to a toluene solvent model (GBSA/GFN2-xTB). (Top row) Histograms showing the distribution of net electron variation, total internal electron density redistribution, and change in dipole moment. (Bottom row) Scatter plot and Pearson correlation matrix illustrating the strong correlation between internal charge redistribution and the change in dipole moment, a key indicator of CT state stabilization by the solvent.

Global property correlations and dimensionality reduction

To uncover the fundamental variables governing TADF properties, Principal Component Analysis (PCA) was employed to distill the high-dimensional property space of the 747 TADF emitters into its most significant underlying variables. The analysis incorporated the following calculated properties: singlet-triplet energy gaps (ΔE_{ST}) from both sTDA and sTD-DFT methods, emission wavelengths (λ_{PL}), oscillator strengths (f_{12}), HOMO-LUMO energy gaps, HOMO-LUMO spatial overlap, donor-acceptor torsional angles, and dipole moments. The analysis reveals a remarkably low-dimensional design space, as detailed in Table 9 and Table 10. In both gas and toluene phases, the first three principal components

(PCs) collectively capture approximately 88 % of the total variance. The dominance of these few components is significant: the first two alone account for over 68 % of the variance (Figure 9), confirming that the vast majority of photophysical behavior is governed by a limited set of orthogonal factors.

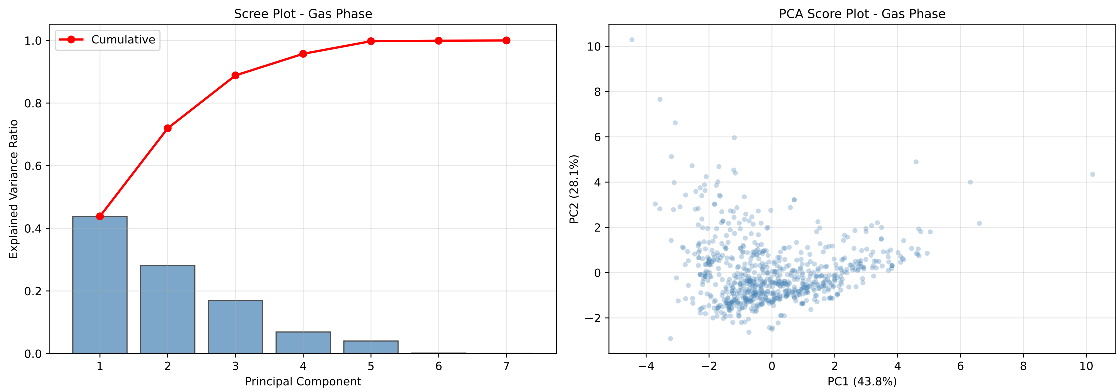
This low-dimensional structure is a consequence of strong underlying correlations between the calculated properties, which are visualized directly in the correlation heatmaps shown in Figure 10 and Figure 11. These maps provide a detailed visualization of the pairwise relationships that underpin the principal components. Several key chemical and physical trends are immediately apparent. For instance, a strong negative correlation is observed between the D-A torsional angle and HOMO-LUMO overlap, quantitatively confirming that twisting the molecular backbone is an effective strategy for spatially separating the frontier orbitals. The expected inverse relationship between $\Delta E_{\mathrm{ST}}$and λ_{PL} is also clearly visible. A comparison between the gas phase and toluene simulations shows a general attenuation of correlation strengths in the solvent phase, indicating that the polarizable environment modulates these intrinsic electronic relationships. Together, the PCA and heatmap analyses confirm that the complex, multi-parameter challenge of TADF emitter design can be rationalized by focusing on a few key properties related to energy gaps, charge separation, and radiative coupling efficiency.

Table 9: Principal Component Analysis (PCA) of the calculated properties for 747 TADF emitters in the **gas phase**. The first three components capture 88.8 % of the total variance.

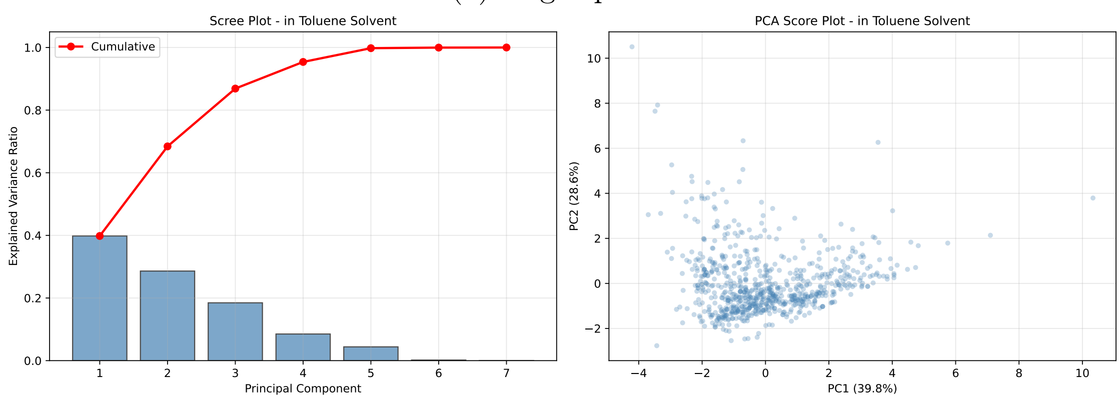
PC	Explained Variance	Cumulative Variance	% Total
PC1	0.438 2	0.438 2	43.8
PC2	0.281 3	0.719 4	71.9
PC3	0.168 9	0.888 3	88.8

Table 10: Principal Component Analysis (PCA) of the calculated properties for 747 TADF emitters in **toluene solvent**. The first three components capture 86.8 % of the total variance.

PC	Explained Variance	Cumulative Variance	% Total
PC1	0.398 1	0.398 1	39.8
PC2	0.286 1	0.684 2	68.4
PC3	0.184 6	0.868 8	86.8



(a) In gas phase



(b) In toluene solvent

Figure 9: Principal Component Analysis (PCA) score plots for the 747 TADF emitters. **Panel (a)** shows the distribution of molecules in the gas phase along the first two principal components (PC1 and PC2), which capture 43.8 % and 28.1 % of the variance, respectively. **Panel (b)** shows the corresponding distribution in toluene, with PC1 and PC2 accounting for 39.8 % and 28.6 % of the variance. In both cases, the data points form a single, dense cluster, illustrating the low intrinsic dimensionality of the TADF property space. The slightly broader distribution in the toluene phase (b) reflects the solvent-induced modulation of molecular properties, but the overall structure of the design space is preserved.

Structure-property relationships: architecture and torsional angles

Having established the reliability and efficiency of our protocol for relative ranking, we leveraged the 747-molecule dataset to extract statistically robust structure-property relationships.

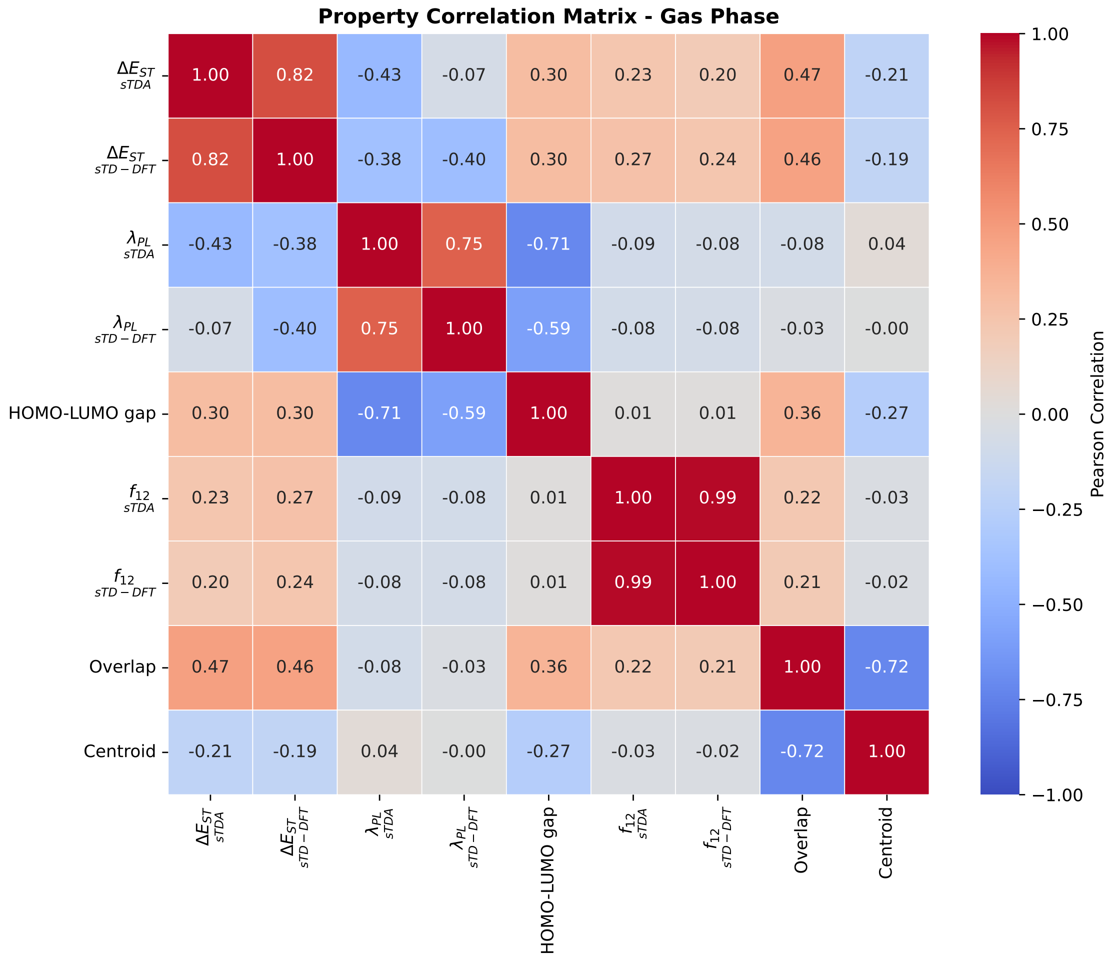


Figure 10: Correlation heatmap of key TADF properties for 747 molecules in the gas phase. The color scale denotes the Pearson correlation coefficient. The map highlights several key relationships: (i) the strong positive correlation ($r \approx 0.82$) between ΔE_{ST}values predicted by sTDA and sTD-DFT, confirming their internal consistency; (ii) the strong negative correlation between torsional angle and HOMO-LUMO overlap, validating the geometric basis for electronic decoupling; and (iii) the moderate negative correlation between $\Delta E_{\mathrm{ST}}$and λ_{PL}, reflecting the fundamental energy gap law.

Our analysis focused on two key structural levers: the overall molecular architecture and the donor-acceptor (D-A) torsional geometry.

First, a clear hierarchy of performance emerges when molecules are categorized by their architecture (Table 11 and Figure 12). Donor–Acceptor–Donor (D-A-D) systems demonstrate statistically superior potential, exhibiting a mean ΔE_{ST} of 0.304 eV, which is significantly lower than that of simple D-A (0.369 eV) or pure donor systems (>0.4 eV). This confirms

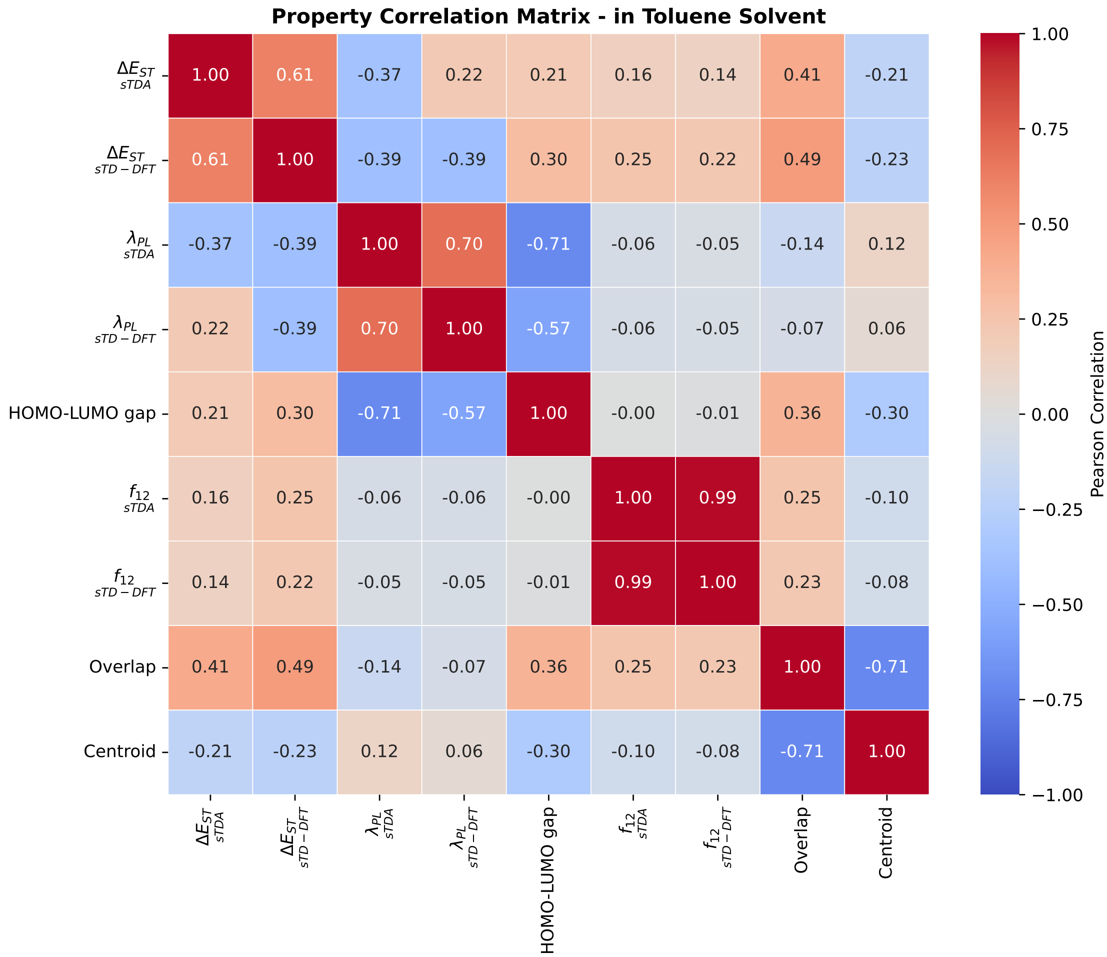


Figure 11: Correlation heatmap of key TADF properties for 747 molecules in toluene solvent. When compared with the gas phase results (Figure 10), this map reveals a general attenuation of correlation strengths (e.g., the sTDA vs sTD-DFT ΔE_{ST}correlation drops to $r \approx 0.61$). This indicates that the polarizable solvent environment introduces additional complexity and modulates the intrinsic electronic and photophysical property relationships observed in vacuum.

that architectural designs promoting charge delocalization across multiple donor sites are highly effective at minimizing the exchange integral, a cornerstone of TADF theory.[6,24]

Second, our large-scale analysis provides statistical validation for the widely held 'design rule' that an optimal D-A torsional angle exists. The torsional angle distribution across our dataset spans a continuous range from near-planar (<10°) to highly twisted (>90°) configurations. To test the optimal angle hypothesis, we grouped molecules into two categories: those within the proposed optimal range (50°–90°) and those outside this range. This binary

Table 11: TADF performance stratified by molecular architecture. The mean $\Delta E_{\mathrm{ST}}$and standard deviation highlight the superior and more consistent performance of D-A-D systems.

Architecture	Count	Mean ΔE_{ST}[eV]	Std Dev
D-A-D	38	0.304	0.129
D-A	20	0.369	0.128
Multi-D/A	181	0.376	0.141
Pure Donor Systems	272	0.425	0.167

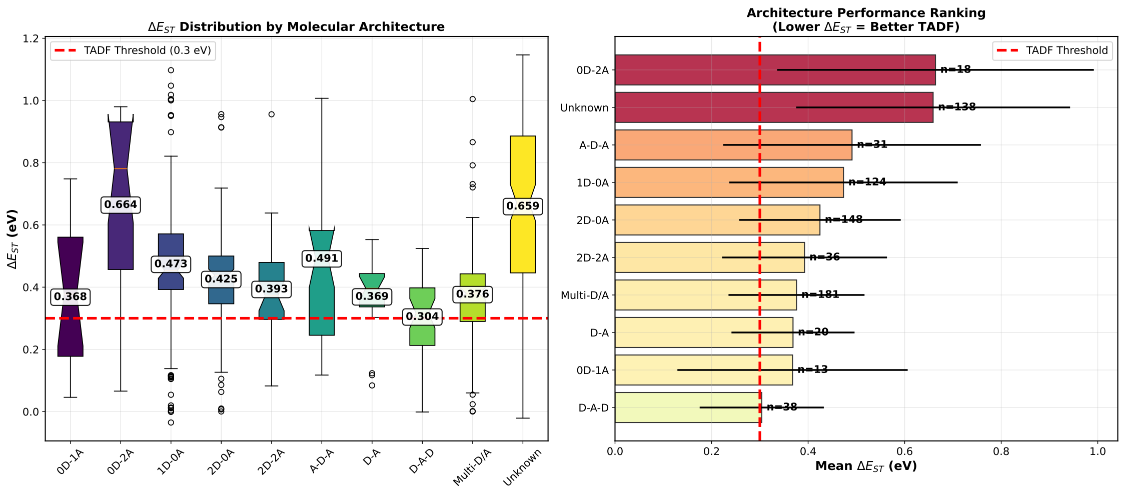


Figure 12: TADF performance stratified by molecular architecture. D-A-D and simple D-A architectures demonstrate superior characteristics (lower mean ΔE_{ST}) compared to systems lacking a distinct acceptor, confirming the efficacy of the donor-acceptor design principle.

grouping was chosen to clearly distinguish between optimal and suboptimal configurations based on established TADF design principles, where angles below 50° typically exhibit insufficient HOMO-LUMO separation and angles above 90° risk excessive decoupling. As shown in Figure 13, molecules with torsional angles between 50° and 90° have a 4.2 percentage point higher probability of being efficient TADF emitters (defined as $\Delta E_{\mathrm{ST}} < 0.3\,\mathrm{eV}$) compared to those outside this range—a statistically significant improvement ($p < 0.001$). This optimal range represents a critical compromise: the twist is large enough to enforce HOMO-LUMO separation and minimize the exchange energy, but not so extreme as to completely decouple the donor and acceptor, which would diminish the oscillator strength.

However, the correlation between torsional angle alone and $\Delta E_{\mathrm{ST}}$is weak across the entire dataset (Figure 14). This indicates that while torsional geometry is a critical tuning

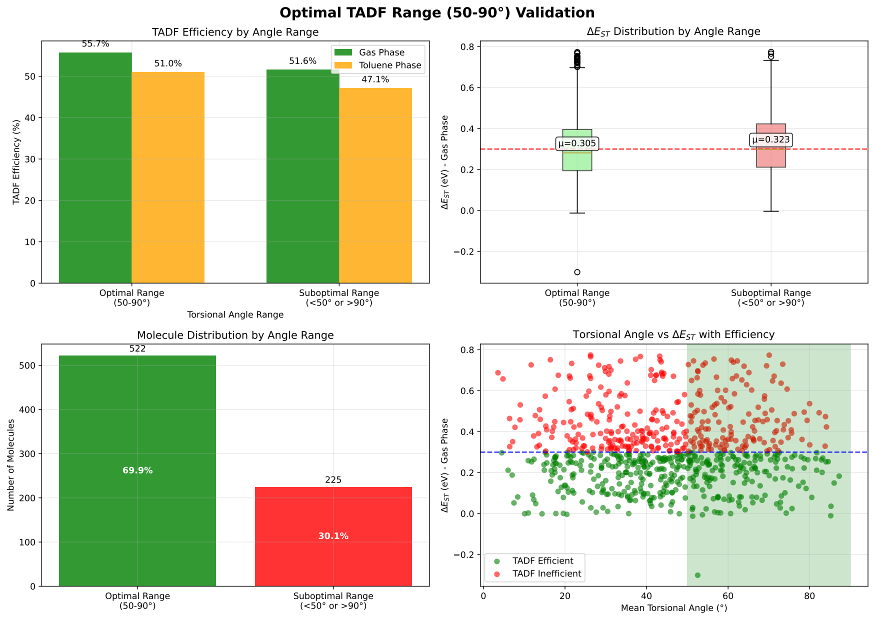


Figure 13: Validation of the optimal torsional angle design rule. The bar chart shows the percentage of molecules that are efficient TADF emitters ($\Delta E_{\mathrm{ST}} < 0.3\,\mathrm{eV}$) within the optimal (50°–90°) versus suboptimal torsional angle ranges. Molecules in the optimal range exhibit a statistically significant increase in TADF efficiency.

parameter, the overall molecular architecture is the primary determinant of TADF potential. For instance, the superior D-A-D architecture performs well across a broad range of torsional angles, underscoring its robustness. This highlights that the most effective design strategy involves first selecting a high-performance architecture (e.g., D-A-D) and then fine-tuning its properties through geometric modifications that enforce an optimal torsional angle.

Conclusions

This study benchmarks semi-empirical xTB methods on 747 TADF emitters—substantially larger than previous validation sets. The protocol we establish is validated for high-throughput screening of TADF candidates. By applying a hybrid computational protocol to 747 experimentally known molecules, we have moved beyond case-by-case analysis to extract statistically robust, quantitative design principles.

Our key findings can be summarized as follows:

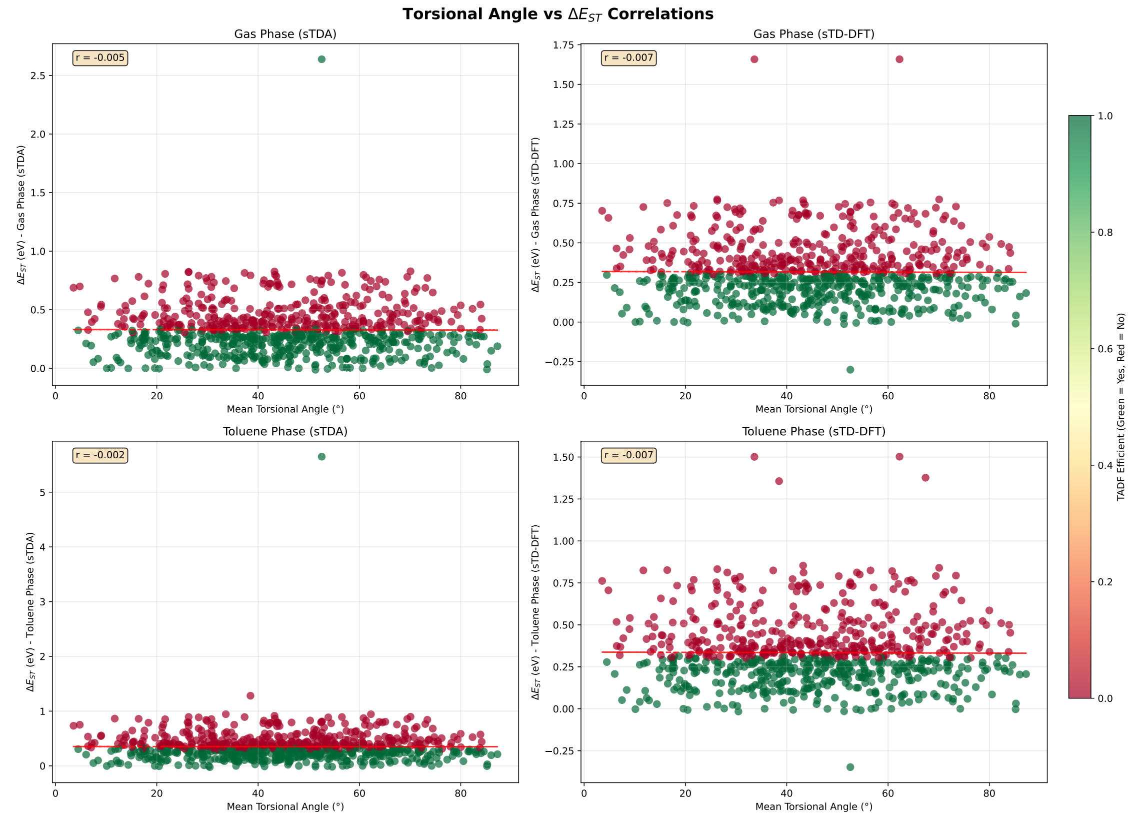


Figure 14: Correlation between D-A torsional angles and the calculated $\Delta E_{\mathrm{ST}}$for all 747 molecules. The lack of a strong linear trend (Pearson $r \approx -0.007$) illustrates that while an optimal range for the torsional angle exists, it is not the sole determinant of the singlet-triplet gap. Molecular architecture and electronic factors play a dominant, confounding role.

- **Methodological validation.** The sTDA-xTB and sTD-DFT-xTB methods are validated as reliable and computationally efficient tools for the relative ranking of TADF candidates. They exhibit strong internal consistency (MAE for $\Delta E_{\mathrm{ST}}$$< 0.03$ eV between methods) and achieve a transformative cost reduction of over 99 % compared to conventional TD-DFT, making large-scale screening feasible on modest computational resources.

- **Predictive accuracy and limitations.** While the methods successfully capture qualitative trends, their quantitative accuracy for absolute prediction is limited by the inherent approximations of the HTS protocol, particularly the vertical approximation. The mean absolute error against experimental ΔE_{ST}values is approximately 0.17 eV, reinforcing that the primary strength of this framework lies in screening and trend analysis, not in yielding quantitatively precise spectroscopic data.

- **Data-driven design rules.** Our large-scale analysis provides statistical validation for key design principles. We confirm that D-A-D architectures are statistically superior to simple D-A and other motifs. Additionally, we identify an optimal D-A torsional angle window of 50°–90° that effectively balances the competing requirements of small ΔE_{ST} and high oscillator strength.

- **Low-dimensional design space.** Principal Component Analysis demonstrates that the vast chemical diversity of TADF emitters can be described by a low-dimensional property space, where nearly 90 % of the variance is captured by just three principal components. This suggests that the design challenge is highly constrained and can be effectively navigated by optimizing a few key orthogonal properties.

The broader impact of this study is the provision of a scalable and validated protocol that bridges the gap between slow, high-accuracy methods and the practical need to explore vast chemical spaces. This work not only provides essential benchmarking data and methodological guidelines for the community but also identifies a curated list of high-priority molecular candidates for future experimental synthesis.

Future work should build upon this foundation. The immediate next step is the experimental validation of the most promising candidates identified herein. From a theoretical perspective, future refinements should focus on developing more accurate yet efficient models for solvation (e.g., state-specific approaches) and incorporating explicit calculations of spin-orbit coupling on representative subsets to refine the mechanistic understanding of the RISC process. Finally, the extensive and well-characterized dataset generated in this study provides an ideal training ground for developing machine learning models capable of further accelerating the discovery of novel, high-performance TADF materials.

The validated protocol, design rules, and candidate list presented here provide practical tools for TADF discovery. We hope these resources prove useful to experimentalists seeking new emitter architectures.

Author Contributions

J.-P.T.N.: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization.
E.V.K.T.: Writing – Review & Editing.
A.M.: Writing – Review & Editing.
S.G.N.E.: Supervision, Writing – Review & Editing.

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Data and Software Availability

The computational dataset of 747 TADF emitters, including optimized geometries, excited-state properties, and analysis scripts, is freely available at [Zenodo DOI: 10.5281/zenodo.17436069] / [GitHub: `https://github.com/TchapetNjafa/Result_article1_TADF_xTB/tree/main`]. All calculations used the xtb package (version 6.7.1) available at `https:`

//github.com/grimme-lab/xtb, the CREST package (version 3.0.2) available at https://github.com/crest-lab/crest, the xtb4stda package (version 1.0) available at https://github.com/grimme-lab/xtb4stda, the stda package (version 1.6.3) available at https://github.com/grimme-lab/std2 and Multiwfn package (version 3.8(dev)) available at http://sobereva.com/multiwfn.

Conflicts of Interest

The authors declare no competing financial interests.

Supporting Information

Stokes shift approximation methodology for estimating relaxed singlet-triplet energy gaps; singlet-triplet correlation plot including Spiro-CN outlier (Figure S1); complete molecular architecture table with SMILES representations for all 747 molecules (Table S1); detailed emission wavelength validation data (Table S2, 213 molecules) including experimental λ_{PL} values, measurement conditions, data type classification (experimental vs computational reference), and literature citations; singlet-triplet gap validation data (Table S3, 296 molecules) with experimental ΔE_{ST} values, measurement methods, data type indicators, and literature references; systematic outlier analysis (Table S4, 20 molecules) comprising 12 emission wavelength outliers (|residual| $>$ 130 nm, MAD-based threshold) and 8 singlet-triplet gap outliers (|residual| $>$ 0.32 eV), each with experimental values, predicted values, residuals, structural classifications (carbazole-based, acridine/phenoxazine, triazine acceptor, polycyclic acceptor, heteroatom-containing), and literature citations; TD-DFT benchmarking study (Table S5, 8 representative molecules) presenting ωB97X-D/def2-TZVP calculations (ORCA 6.1.0) including S_1 and T_1 energies, ΔE_{ST} values, oscillator strengths, comparison with sTDA-xTB and sTD-DFT-xTB predictions, systematic deviation analysis ($MAE \approx 0.77\,\mathrm{eV} - -0.79\,\mathrm{eV}$), and validation of relative ranking preservation.

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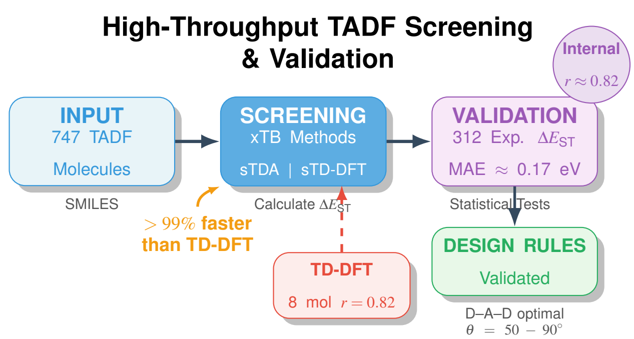

Figure 15: TOC Graphic